

SF Planning Commission Minutes: Pipeline Status and Preliminary Results

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1 Purpose

This directory documents the empirical backbone of the project: turning twenty-eight years of San Francisco Planning Commission meeting minutes (1998–2026) into an *item-level* dataset of discretionary land-use decisions. One row is one request heard by the Commission — a conditional use, a discretionary review, a variance, a rezoning — with its case number, address, zoning district, staff recommendation, disposition, roll-call vote, and hearing date.

That dataset is what makes the “market for housing regulation” measurable. It lets us ask who obtains discretionary approval, how often staff recommendations are overridden, how the approval rate moves with the political composition of the Commission, and how any of this responds to state preemption. The theory side lives in `output/political_economic_housing_model/`; the extension of the same pipeline to the rest of the Bay Area is documented in the recon directories.

2 What is in this directory

The directory is **one folder per memo**. Each holds `<name>.tex`, its compiled `.pdf`, and its own `figures/` and `tables/`, so a memo and everything it depends on move together; every figure and table is written by a script, and each memo’s last section names the script that regenerates it. `README.md` in this directory is the index. The spec and reference documents the pipeline code links to by path live in `notes/`.

Folder	What it answers
<code>memo/</code>	This memo: pipeline status and what is still open. Edited in place.
<code>meeting_level_info/</code>	How each item is matched to the meeting it was heard at, the meeting-level fields, and their accuracy against the gold set.
<code>extraction_method_comparison/</code>	Regex against Claude, field by field, on a frozen train/test split, with accuracy and over-extraction reported separately.
<code>discretionary_review_patterns/</code>	What the 16,199-item corpus contains: composition of the docket, dispositions, delay, commissioners, geography, Planning Code citations.
<code>permit_linkage/</code>	How far the entitlement can be followed to a real building permit in DBI’s records, and what the match buys.

Folder	What it answers
<code>conditions_of_approval/</code>	Whether the <i>content</i> of the Commission’s conditions can be recovered, given that <code>conditions_imposed</code> is only a yes/no flag.
<code>predicting_delay/</code>	How much elapsed hearing time is forecastable from what a developer can observe at the first hearing — in sample and out.
<code>notes/</code>	<code>labeling_rules.md</code> (the coding manual), <code>data_infrastructure.md</code> (schema and flow), <code>minutes_data_availability.md</code> , <code>processing_review.md</code> , <code>schema_enrichment_recommendation.md</code> , <code>hand_label_review_guide.md</code> , <code>help_string_audit.md</code> .

3 The pipeline

Scrape → **parse** → **date** → **label** → **extract**. Code lives in `code/commission_minutes_processing/`; the corpus lives outside the repo on Dropbox and is reached through `paths.py`.

Parsing splits each source document into blocks, one per agenda item, delimited by case headers, numbered agenda items, section headers, adjournment markers, and draft-minutes-adoption appendages. Two document eras are handled: scraped HTML (1998–2014) and PDF (2015–present).

Dating is a separate stage, and deliberately so. A block never states its own hearing date — the meeting header above it does — so the date is a structural question about which meeting a block falls under, not a textual one about the block. It is answered by `assign_meeting_dates.py` and validated against hand-marked meeting boundaries collected in `date_boundary_app/`. See the companion memo, `meeting_level_info.tex`.

Meeting-level labelling sits above the items. Time of day, whether the sitting was regular, special, joint or closed, the room, who sat and who staffed it are properties of the hearing, shared by every item heard at it, so they are recorded once per meeting and joined on the date rather than repeated on 23,000 rows. `meeting_headers.py` cuts the header window around each known meeting start and pre-fills twelve such fields for confirmation in the boundary app.

Labeling is done in a local Flask app against the shared schema, with a machine pre-fill the human corrects. *Extraction* then either fine-tunes a model on those labels or few-shot prompts one, and `learning_curve.py` plots field accuracy against label count so labeling stops at the empirical plateau rather than a guess.

3.1 The schema is the single source of truth

`extraction_common.py` holds one list, `SCHEMA`, from which the labeling form, the model prompt, the training builder’s required keys, and the evaluation metric are all generated. Editing that list propagates everywhere. It currently carries **28 fields** in six groups: Identity, Location, Zoning & scale, Process, Politics, Conditions.

Four fields have been dropped as the schema settled, each for a stated reason: `jurisdiction` (constant), `supervisory_district` (recoverable from the address), `item` (agenda number, no

analytic signal), and — as of 30 August 2026 — `meeting_date`, which is a property of the meeting rather than the item and is now supplied by the dating stage.

Two design decisions in the schema are worth stating because they carry the empirics. First, `action` records the disposition *family* (`approved`, `disapproved`, `continued`, `took_dr_and_approved`, ...), with “with conditions” and “as modified” split out into two orthogonal flags. This removed a three-way overlap in the old vocabulary and lets conditions attach to non-approval outcomes. Second, the staff recommendation is stored twice: verbatim (`preliminary_recommendation`) and bucketed to the same disposition families (`preliminary_recommendation_category`). The gap between that category and `action` is the staff-override signal the project studies, and storing both makes “was the staff recommendation followed?” a direct enum comparison rather than string wrangling.

4 Preliminary results

4.1 Corpus

Quantity	Count
Parsed items, 1998–2026	23,043
HTML era (1998–2014)	16,024 across 724 pages
Modern era (2015–2026)	7,019
Meetings identified, HTML era	818
Pages holding more than one meeting	41 (max 5)

4.2 Dating

The dating stage corrected **1,937 of 16,024** HTML-era item dates. Distinct hearing dates per year moved from 12–17 to 35–47 for 1998–2000, in line with the Commission’s roughly weekly sitting; later years were already close to correct. Against a hand-marked gold standard of 97 documents and 101 meeting boundaries, boundary precision and recall are both 1.000 and item date accuracy is 1.000 on 1,322 scored items — and on the 21 documents marked after the rules were frozen it is 1.000 on 334 items, so the result is genuinely out-of-sample. All 280 modern PDFs agree with their file names. Details and caveats are in the companion memo.

4.3 Meetings

All **102 meetings** of the gold sample are hand-confirmed — the 101 marked boundaries plus one the dating stage found that the marking missed. Against those labels the ten meeting-level fields agree on **94.6%** of values, and on the 21 meetings labelled after the rules were frozen they agreed on **92.1%** before correction, so the extraction generalises. The sample is already informative about the panel’s composition: 90 regular sittings, 7 special, and 5 joint — with the Redevelopment Agency Commission (1998), Building Inspection (2007, 2018), Public Health (2011) and Historic Preservation (2023). Two joint sittings are headed simply “Special Meeting” and are identifiable as joint only from the two bodies named above the date.

Status	Items	Meaning
done	119	confirmed gold
review	80	existing labels with a specific QA issue to recheck
flagged	179	145 representativeness-sample draws + 34 recovered labels
todo	22,665	unlabeled

4.4 Labeling

The queue is ordered rare-class-first and interleaved by year, so labeling top-down fills scarce request types and dispositions evenly across the panel rather than exhausting 1998 first. A fixed random sample per year is flagged separately so the gold set is representative of the whole period rather than of whichever years were labeled earliest.

5 Status and open questions

- **Settled.** The schema; the parser’s block boundaries; the dating stage and its validation; the labeling workflow and coding manual.
- **In progress.** Item-level hand-labeling. The immediate queue is the 80 **review** items, then the 145 representativeness draws. The meeting-level gold set (97 documents, 102 meetings) is complete.
- **Open.** Forty-two labels carry a hand-typed date that disagreed with the dating pipeline (preserved in `date_field_disagreements.csv`); several look like labels attached to the wrong block by the earlier content-matching recovery and need a separate pass. The learning curve has not yet been run at full size, so the number of labels required for the extraction model is still an estimate rather than a measurement.
- **Decision pending.** Whether to extend meeting-level labelling beyond the 81-meeting pilot to all 818 detected meetings. The pre-fill is good enough that the cost is confirmation rather than transcription, but the analytic payoff of the meeting-level fields is not yet established.
- **Known non-transfer.** The heuristic extractor is tuned to SF minutes and does not transfer to other jurisdictions; extending the pipeline requires re-testing extraction on real documents with human review, not unattended.